15-Second AI Video Challenge

Theme: Marwadi University in 2050

Field	Details
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Department	ICT
Batch	7EK1'A'
Course	Prompt Engineering for Generative AI

1. Final Video Link

<<< PASTE YOUR PUBLIC GOOGLE DRIVE VIDEO LINK HERE >>>

Duration 15.00 seconds · 576×320 · 8 fps · 120 frames. Upload marwadi_2050_15s.mp4 to Google Drive, set sharing to “Anyone with the link”, and replace the line above before exporting this document to PDF.

2. Video Generation Model

Item	Details
Model	cerspense/zeroscope_v2_576w
Type	Open-source text-to-video latent diffusion
Base	Fine-tuned from ModelScope text-to-video
Why chosen	Watermark-free, clean 576×320 output, and fits a free Colab T4. A larger model such as CogVideoX gives better fidelity but risks running out of memory mid-render.
Fallback	damo-vilab/text-to-video-ms-1.7b
Framework	Hugging Face diffusers (DiffusionPipeline)
Hardware	Google Colab, NVIDIA T4

3. Structure of the Video

The video uses the **5 × 3-second clips** option from the brief. Five clips of 24 frames each are generated at 8 fps and combined into one continuous file of exactly 120 frames — 15.00 seconds.

The five shots form a single narrative arc — **arrive → enter → learn → build → graduate** — so the sequence reads as a story rather than five unrelated clips.

Shot	Time	Beat	Camera movement
1	0–3 s	Arrival	aerial drone shot slowly pushing forward and descending toward the main building
2	3–6 s	Entry	low tracking shot moving forward behind the students, steadicam
3	6–9 s	Learning	slow dolly-in toward the hologram, eye level
4	9–12 s	Building	low-angle shot slowly orbiting around a workstation
5	12–15 s	Graduation	crane shot rising upward and tilting to reveal the whole campus skyline

4. Prompts Used

Every prompt is composed from **six explicit control dimensions** — subject, environment, camera, lighting, style and quality — so each shot is directed rather than merely described.

Dimension	Purpose
Subject	Who or what is in frame
Environment	Where, and what the world looks like

Camera	Shot type and its movement
Lighting	Time of day and mood
Style	Visual treatment
Quality	Technical tags

Continuity anchor — repeated verbatim in all five prompts so the five clips read as one place and one film:

```
futuristic university campus, glass and steel architecture with hanging greenery, warm teal and amber colour grade
```

Style and quality tags — appended to every prompt:

```
cinematic film still, photorealistic, shallow depth of field, volumetric light,
highly detailed, sharp focus, smooth natural motion, 4k
```

Negative prompt — applied to every shot:

```
blurry, low quality, distorted, deformed faces, extra limbs, text, watermark, logo, subtitles, jpeg artifacts, flickering, static frame, cartoon, low resolution, oversaturated
```

Full prompt for each shot

Shot 1 — Arrival (0–3 s)

```
a wide view of a grand university campus with students arriving, futuristic university campus, glass and steel architecture with hanging greenery, warm teal and amber colour grade, wide plazas, solar canopies, autonomous shuttles gliding along a tree-lined avenue, aerial drone shot slowly pushing forward and descending toward the main building, golden sunrise light, long soft shadows, light morning haze, cinematic film still, photorealistic, shallow depth of field, volumetric light, highly detailed, sharp focus, smooth natural motion, 4k
```

Shot 2 — Entry (3–6 s)

```
groups of students in smart casual uniforms walking together, carrying transparent tablets, futuristic university campus, glass and steel architecture with hanging greenery, warm teal and amber colour grade, a luminous holographic entrance archway projecting the university crest, low tracking shot moving forward behind the students, steadicam, soft morning daylight with cyan holographic glow on their faces, cinematic film still, photorealistic, shallow depth of field, volumetric light, highly detailed, sharp focus, smooth natural motion, 4k
```

Shot 3 — Learning (6–9 s)

```
students and a professor around a floating three-dimensional hologram of a molecule, futuristic university campus, glass and steel architecture with hanging greenery, warm teal and amber colour grade, a bright future laboratory with robotic arms and curved glass display walls, slow dolly-in toward the hologram, eye level, cool interior lighting with bright cyan light spilling from the hologram, cinematic film still, photorealistic, shallow depth of field, volumetric light, highly detailed, sharp focus, smooth natural motion, 4k
```

Shot 4 — Building (9–12 s)

```
focused students coding at glowing workstations during a late-night hackathon, futuristic university campus, glass and steel architecture with hanging greenery, warm teal and amber colour grade, an open innovation hall at night, monitors and LED strips glowing, city lights beyond the glass, low-angle shot slowly orbiting around a workstation, dark room lit by monitor glow and amber desk lamps, strong rim light, cinematic film still, photorealistic, shallow depth of field, volumetric light, highly detailed, sharp focus, smooth natural motion, 4k
```

Shot 5 — Graduation (12–15 s)

```
graduating students in ceremonial robes throwing their caps into the air, celebrating, futuristic university campus, glass and steel architecture with hanging greenery, warm teal and amber colour grade, a vast open convocation ground with a crowd and a lit stage, camera drones hovering above, crane shot rising upward and tilting to reveal the whole campus skyline, warm golden hour sunset, glowing particles and confetti in the air, cinematic film still, photorealistic, shallow depth of field, volumetric light, highly detailed, sharp focus, smooth natural motion, 4k
```

5. How the Video Was Generated

1. **Model load.** Zeroscope v2 576w is loaded through diffusers with fp16 weights, VAE slicing and model CPU offload so it fits a free T4.

2. **Per-shot generation.** Each of the five prompts is rendered independently to 24 frames at 576×320, using 40 inference steps and guidance scale 12.0. Each shot has its own fixed seed (101, 202, 303, 404, 505), so the whole render is reproducible.

3. **Individual clips saved.** Every shot is written to its own MP4 in clips/ before anything is joined — those files are the evidence that five separate clips were generated.

4. **Combination.** The clips are concatenated into one continuous MP4 (section 6).

5. **Verification.** The notebook asserts the final duration is within the 15-second requirement before finishing.

Parameter	Value
Frames per clip	24
Frame rate	8 fps
Resolution	576×320
Inference steps	40
Guidance scale	12.0
Seeds	101, 202, 303, 404, 505
Cross-dissolve	0 frames per join
Final length	120 frames = 15.00 s

6. How the Clips Were Combined

The five clips are joined by concatenating their frame arrays and writing a single MP4 with imageio — no external editor was used, so the combination step is fully reproducible from the notebook.

The frame arithmetic is the part worth explaining. A cross-dissolve blends frames from both sides of a join rather than adding new ones, so it shortens the result. Five 24-frame clips with a 6-frame dissolve at each of the four joins would give 5×24 − 4×6 = 96 frames = 12 seconds, not 15. Two measures prevent that:

• Frames per clip is derived from the target — ceil((120 + joins × XFADE) / 5) — so enough footage is generated to absorb the dissolves.

• The concatenation is then trimmed to exactly 120 frames.

The submitted version uses XFADE = 0 so the five shots meet at hard cuts, which is the normal convention for a short montage and lands on exactly 120 frames. Raising XFADE produces dissolves instead; the duration stays 15 seconds either way.

```
clip 1  [24 frames]  0.0-3.0s
clip 2  [24 frames]  3.0-6.0s
clip 3  [24 frames]  6.0-9.0s
clip 4  [24 frames]  9.0-12.0s
clip 5  [24 frames]  12.0-15.0s
                |
     concatenate frame arrays (0-frame blend at each join)
                |
     trim to 120 frames  ->  marwadi_2050_15s.mp4  (15.00 s)
```

7. Prompt Engineering — What Actually Controlled the Output

The brief states the objective is not a beautiful video but demonstrable control over scene, characters, environment, camera movement and visual storytelling. These were the changes that produced that control:

#	Problem observed	Change made	Result
1	Clips looked like five unrelated places	Added a shared continuity anchor to every prompt	Reads as one campus
2	Almost no motion — nearly a still image	Gave camera movement its own explicit clause	Consistent directed motion
3	Warped faces in crowd shots	Added deformed faces, extra limbs to the negative prompt	Fewer artefacts
4	Colour jumped between shots	Fixed the colour grade inside the continuity anchor	Consistent grade
5	Text and watermarks appearing	Added text, watermark, logo, subtitles to the	Clean frames

		negative prompt	
6	Weak prompt adherence	Raised guidance scale to 12.0	Shots match direction

The most useful lesson: in text-to-video, camera movement is a prompt component, not an afterthought. Describing a scene yields a nearly static shot; describing a camera moving through the scene is what produces motion. Storytelling here is carried by the camera — push-in, track, dolly, orbit, crane-up — not by cutting between unrelated images.

8. Files Submitted

File	Contents
marwadi_2050_15s.mp4	The final 15-second video
clips/shot1..shot5.mp4	The five individual 3-second clips
storyboard_contact_sheet.png	Middle frame of each shot
video_manifest.json	Every prompt, seed and parameter from the run
Marwadi_2050_Video.ipynb	The notebook that generated everything